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Adversarial Validation: Stress-Testing Data-Driven Discoveries with Automated Critique

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ABSTRACT

We describe a protocol for validating data-driven discoveries in astrophysics using automated adversarial critique. After a candidate model is discovered via symbolic regression, an independent agent reads all code, data, and claims, then systematically challenges every step. The defender responds, concedes honestly where needed, and fixes errors before the next round. Applied across 4 projects involving Hubble constant inference, the radial acceleration relation, black hole scaling laws, and galaxy formation simulations, the protocol has caught 7 distinct classes of error in pre-publication code and interpretation — including duplicate data points, integration bugs, column naming mismatches, degenerate statistical tests, circular mass calibrations, and aspirational survey forecasts passed off as realistic. None of these errors was flagged by standard human review. We argue that automated adversarial validation should become a standard step in data-driven astrophysics, complementary to (not replacing) human review.

1. INTRODUCTION

Data-driven methods — symbolic regression, machine learning, automated model discovery — are transforming astrophysics. They find patterns that humans miss, scale to large datasets, and impose minimal theoretical priors. But they also introduce new failure modes. A symbolic regression search might find a functional form that fits perfectly but extrapolates pathologically. A neural network can memorize noise. A bootstrap can conceal non-quadratic likelihood surfaces.

Standard validation tools — holdout tests, bootstrap, cross-validation — are designed to catch statistical overfitting. They are not designed to catch conceptual errors: mislabeled columns, circular reasoning, units confusion, duplicate data, or aspirational claims dressed as results.

We have developed a complementary validation protocol: adversarial debate. After the analysis is complete, an independent agent reads everything — data files, analysis scripts, and the manuscript — and systematically challenges every claim. The defender must respond point by point, conceding honestly. Errors are fixed, limitations are documented, and the process repeats until zero fatal findings remain.

This paper describes the protocol and presents 7 case studies of errors it has caught across 4 research projects, plus a detailed worked example from a Hubble constant analysis.

2. PROTOCOL

2.1. *Initial Analysis*

The researcher completes the standard analysis pipeline:

1. Parse public data into (X, y) with proper errors, verified against published tables.
2. Run symbolic regression (PySR Cranmer (2023)) with multiple random seeds and model selection by accuracy.
3. Validate the discovered form: bootstrap resampling, holdout tests, cross-sample verification, literature comparison.
4. Extend to independent data and theory predictions.
5. Draft a manuscript reporting results and conclusions.

2.2. *Adversarial Review*

An independent agent — in our implementation, a large language model with file-reading and code-execution capabilities — reads all project files and conducts a systematic audit:

- **Data integrity:** Check every measurement against its published source. Look for duplicates, unit mismatches, and transcription errors.
- **Methodology:** Test sensitivity to arbitrary thresholds, extrapolation beyond the data range, and choices of fitting procedure.

- **Interpretation:** Challenge causal claims, circular reasoning, and "both methods are valid" statements that paper over real discrepancies.

- **Presentation:** Flag overclaims, aspirational language masquerading as results, and missing uncertainties.

2.3. *Defender Response*

The defender responds to each challenge, categorizing it as:

- **Conceded:** The error is real and must be fixed.

- **Partially sustained:** A genuine limitation exists but does not change the central result; it should be acknowledged.

- **Rejected:** The challenge is based on a misunderstanding or is technically incorrect.

Fixes are applied for all conceded and partially sustained challenges. The resolution and any code changes are documented in a running debate log. A subsequent round re-evaluates the fixed analysis. The process terminates when zero fatal findings remain.

2.4. *Adversarial Prompt Template*

The following prompt is used to initialize each adversarial review round. It is issued to a large language model with file-reading and code-execution tools, and the model has access to the full project directory.

```
You are an adversarial reviewer auditing a research project.
Read ALL files: data, code, and the manuscript.
Then systematically challenge every claim you find. Categories:
1) Data integrity: duplicates, unit mismatches, transcription errors.
2) Methodology: sensitivity to thresholds, extrapolation, fitting choices.
3) Interpretation: circular reasoning, overclaims, hidden assumptions.
4) Presentation: missing uncertainties, aspirational language,
   undocumented priors.

For each challenge, cite specific line numbers or file paths.
Be aggressive - your job is to find problems.
The defender will respond; errors will be fixed before the next round.
```

The same prompt is re-used in subsequent rounds with the updated codebase.

3. CASE STUDIES

All case studies come from projects using PySR symbolic regression applied to astrophysical datasets. The adversary was the opencode CLI tool using a large language model backend. In each case, the error had survived standard human review before the adversarial round.

3.1. *Duplicate Data Points (BH M–σ)*

Context: Compilation of 126 supermassive black hole masses with bulge velocity dispersions, drawn from multiple literature sources.

Error: Two independent literature searches contributed overlapping catalogs. The galaxy NGC 4258 appeared twice — once from an H_2O maser measurement and once from a stellar-kinematic measurement — with different mass values (3.9 and $3.6 \times 10^7\,M_\odot$) treated as separate objects. Three other galaxies had similar duplicate entries.

Resolution: Deduplication logic added. The duplicate entries accounted for 4 of the 126 points (3% contamination), which shifted the best-fit slope by 0.07 ± 0.04, within uncertainty but non-negligible.

3.2. *Integration Accuracy (H_0 Cosmology)*

Context: Symbolic regression discovered an $H(z)$ polynomial from cosmic chronometer + BAO data. Supernova distance moduli required numerical integration $\int_0^z dz'/H(z')$.

Error: The initial implementation used a Riemann sum with $n = 2000$ points at fixed redshift spacing. At low z (< 0.05), the $1/H(z')$ integrand varies rapidly, and 2000 points undersampled the first ∼10 bins. Error reached 0.24 mag at $z = 0.01$, adding a 7 km/s/Mpc bias to H_0.

Resolution: Switched to Simpson's rule with adaptive spacing. Verified convergence at $n = 5000$ to $\Delta\mu < 10^{-14}$ mag. The 7 km/s/Mpc bias was measured by re-running the full H_0 fit with both integration methods and comparing the resulting H_0 values.

3.3. *Degenerate Statistical Null (RAR Hooks)*

Context: Search for non-monotonic "hook" features in the radial acceleration relation (RAR) of 171 SPARC galaxies.

Error: The null hypothesis for hook significance was a parametric fit (CPX5 form). For 28 of 171 galaxies, the CPX5 fit was degenerate ($\chi^2_{\rm red} > 5$), falsely inflating apparent hook significance. The parametric null was underfit to the data, making any residual look significant.

Resolution: Replaced the parametric null with a Gaussian Process (RBF kernel), which flexibly fits any smooth curve. Under the GP null, 3 of 164 galaxies showed significant hooks — consistent with the expected 5% false-positive rate ($p = 0.99$ binomial). The hook population was eliminated.

3.4. *Column Name Mismatch (RAR)*

Context: Cross-matching simulated and observed galaxy catalogs for RAR comparison.

Error: The simulation column labeled `halfmassrad` was loaded and analyzed as the half-mass radius. In the data pipeline, `vmaxrad` (radius of maximum velocity, typically 2–3× larger) had been used instead for the SPARC sample. The mismatch propagated through 3 analysis scripts before the adversary flagged it.

Resolution: Standardized on a single definition. The original SPARC-based result was unaffected (it used `vmaxrad` consistently), but the simulation comparison had to be re-run with the correct definition.

3.5. *Wrong Physical Variable (Fundamental Plane)*

Context: Testing whether a SR-discovered galaxy scaling relation could replace the fundamental plane for distance estimation.

Error: The candidate form used M_*/R_e^2 (stellar mass surface density) as a proxy for surface brightness. These quantities are correlated but not interchangeable — M_*/R_e^2 includes the mass-to-light ratio, which varies systematically with galaxy type. The candidate was predicting M_*/L variation, not a fundamental plane.

Resolution: The candidate was reframed as a mass-based scaling relation rather than a fundamental plane replacement. The physical interpretation changed accordingly.

3.6. *Circular Mass Calibration (BH M–σ)*

Context: Compilation of 126 BH masses from two methods — 96 from stellar/gas kinematics (direct) and 30 from reverberation mapping (RM). RM masses are calibrated to the M–σ relation itself.

Error: The RM points were included to increase sample size, but their masses are not independent measurements — they assume the M–σ relation that the analysis was attempting to measure. Including them in the fit biases the slope toward the input calibration.

Resolution: RM points flagged as not independent. Primary result reported from kinematic-only sample (96 points); RM-inclusive fit reported separately as a consistency check.

3.7. *Aspirational Forecast Factors (RAR)*

Context: Forecast of how quickly future surveys (Euclid, Rubin, Roman) could distinguish competing dark matter and modified gravity models.

Error: The forecast used the maximum theoretical galaxy counts from survey science requirement documents (SRDs), not realistic yields accounting for target allocation, weather loss, and processing cuts. For Euclid, the SRD cites 1.5×10^9 galaxies; realistic yields after selection cuts are $2\text{–}4 \times 10^8$. The factor-4 overestimate propagated to optimistic timelines.

Resolution: Replaced SRD maximum counts with realistic yields from survey operating plans. For example, Euclid's predicted year for distinguishing dark matter models from modified gravity shifted from year 5 to year 7 of survey operations — a 2–3 year delay.

4. WORKED EXAMPLE: HUBBLE CONSTANT

We provide a detailed walkthrough of the adversarial protocol applied to a joint analysis of cosmic chronometers, BAO, DESI DR2, and Pantheon+ SNe, discovering $H(z) = H_0 + A\,z\,(z - B)(z^2 + C)$ with $f(0) = 0$.

The adversary raised 14 challenges across two rounds:

ROUND 1 (7 CHALLENGES)

- **r$_d$ marginalization:** BAO r_s conversion uses a Planck prior $[146, 148]$ Mpc, which does not test model-independent $r_s \in [130, 160]$ Mpc. *Sustained:* Noted as a limitation; the no-DESI test (CC+SDSS only, $H_0 = 67.0$) provides the model-independent check.
- **H(z)-only vs joint:** $H(z)$-only fit gives $H_0 = 65.4$, 2.3σ below Planck. *Sustained:* Reflects weak extrapolation of the polynomial from $z > 0.07$ to $z = 0$; the SNe anchor the full joint result.
- **Sample-dependent H_0:** Three SN samples give H_0 spanning 66.4–68.3 (~ 1.9 km/s spread). *Sustained:* Acknowledged as a systematic floor.
- **Reduced χ^2:** $\chi^2_{\mathrm{SN}}/N = 0.88$ suggests overestimated errors. *Rejected:* Conservative systematics are standard; the relative $\Delta\chi^2$ (fix-M test) is unaffected.
- **BAO r_s circularity:** Converting D_H/r_s to $H(z)$ assumes r_s. *Partially sustained:* r_s marginalized with Planck prior; no-DESI test is r_s-independent.
- **CC survey heterogeneity:** 33 points from 6 surveys. *Partially sustained:* No cross-correlation expected, but noted.
- **Duplicate $z = 0.51$:** SDSS and DESI both contribute at $z = 0.51$. *Partially sustained:* Independent surveys; consistency verified ($\Delta = 2.2 \pm 2.8$, $< 1\sigma$).

ROUND 2 (7 COUNTER-ATTACKS)

The second round escalated to deeper methodological challenges:

- **Planck-centric r$_d$ grid** — sustained, limitation noted.

- **Fix-M test symmetry** — the test identifies inconsistency but not culprit; symmetry is broken by three-sample corroboration.
- **Bootstrap vs profile discrepancy** — $4\times$ uncertainty difference explained by conditional vs. marginal inference.
- **Conservative covariance** — does not affect $\Delta\chi^2$.
- **Coarse M(z) grid** — adequate for $< 2\sigma$ null, not precision.
- **DR1 vs DR2 mismatch in ΛCDM comparison** — *conceded:* ΛCDM fit used DR1 while SR used DR2; re-run with DR2 gave consistent results.
- **Duplicate paragraph in manuscript** — *conceded:* copy-paste error removed.

OUTCOME

Zero fatal findings across 14 challenges. 3 conceded (technical errors), 10 partially sustained (acknowledged limitations), 1 rejected. The core result ($H_0 = 68.0 \pm 0.9$, $> 6\sigma$ SH0ES exclusion) survived unchanged.

5. DISCUSSION

5.1. *What the protocol catches*

All seven case studies come from symbolic regression (PySR) projects. The protocol has not yet been tested on neural networks, gradient-boosted trees, or other machine learning pipelines. We expect the same categories of error to appear (data integrity, circular reasoning, aspirational forecasts) but caution that the demonstrated scope is SR-based discovery.

Across 4 projects, the 7 case studies above reveal a pattern: the errors caught by adversarial review are *conceptual*, not statistical. None would have been flagged by a bootstrap or cross-validation. They arise from:

- **Propagation of initial choices** — one wrong column name, one wrong integration method, one duplicated catalog entry silently corrupts downstream results.
- **Circular reasoning** — including calibration-dependent data in a fit that measures the calibration.
- **Aspirational vs realistic** — forecasts based on instrument design limits rather than operating reality.
- **Presentation errors** — duplicate text, overclaimed significance, undocumented priors.

5.2. *What it does not catch*

The protocol is complementary to human review, not a replacement:

- It cannot assess physical plausibility beyond what it has been trained on.
- It misses errors that are invisible in the code and data — e.g., a systematically wrong telescope calibration.
- It is only as thorough as the adversary's instructions. A lazy adversary ("looks fine to me") is worse than none.

5.3. *Circularity and Independence*

Because the same LLM-based tool is used for code development, manuscript writing, and adversarial review, a concern arises: is the AI checking its own work? In our implementation, the adversary is invoked as a separate session with a fresh context and explicit adversarial instructions (Section 2.4). It has access to all files but no memory of how the analysis was constructed. This independence is imperfect — a different LLM instance with the same underlying training data — but the protocol detected genuine errors even in AI-written code and text (e.g., the duplicate paragraph in Section 4). Users should be aware of the residual circularity and may prefer a different model for the adversarial role.

5.4. *Cost and Practicality*

Each adversarial round costs approximately \$2–5 in API tokens (GPT-4-class model) and requires 30–60 minutes of wall-clock time. The defender response takes 1–2 hours per round. A typical project completes in 2–3 rounds. Of the challenges raised, approximately 40% are rejected (adversary misunderstanding), 50% are partially sustained (genuine limitation, central result unaffected), and 10% are conceded (actual error requiring a fix). The false-positive rate is high, but the cost per false alarm is low — a few minutes of human review.

5.5. *Recommendations*

1. **Make it standard:** Adversarial validation should be a standard step in data-driven astrophysics, as routine as bootstrap or cross-validation.
2. **Document the debate:** A debate log with every challenge and response should accompany the paper as supplementary material.
3. **Release the protocol:** We provide a template in our repository (https://github.com/ivan-hernandez/h0-symbolic-regression).

4. **Be honest:** Concede limitations freely. A paper that acknowledges 10 non-fatal limitations is more credible than one that claims 0.

6. CONCLUSION

Automated adversarial validation has caught 7 distinct classes of error across 4 projects, none of which was flagged by standard human review. The protocol is simple, reproducible, and complementary to existing validation methods. We recommend it as a standard step in data-driven astrophysics research, particularly for projects using symbolic regression, machine learning, or other automated discovery techniques where the conceptual failure modes differ from traditional statistical overfitting.

This research made use of PySR Cranmer (2023), NumPy Harris et al. (2020), Astropy Astropy (2018), and public data from the Pantheon+, DES-SN5YR, DESI, and SPARC collaborations. AI assistance (opencode) was used for code development, adversarial review, and manuscript preparation.

REFERENCES

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